Downsampling & Upsampling Analysis
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1. **Experiment Overview**

   In this experiment, I tested how different downsampling and upsampling methods affect digital image. I used 3 images with different characteristics. 1 image with a lot of edges & text, 1 natural scene, & 1 image with a lot of fine details such as leaves and branches.

   For the downsampling part, I tested mac pooling, average pooling, & median pooling. I first used factor of 2, but since the differences were still quite hard to see, I also tested factor 4! And the effect of each method became much more noticeable.

   For the upsampling part, I used the images that had already been downsampled using average pooling with factor 4. Then I enlarged them again using nearest neighbor, bilinear interpolation, & bicubic interpolation so I could compare how each method reconstructs the image.

2. **Downsampling analysis**

   The first thing I noticed was how much the downsampling factor matters. With factor 2, max, average, & median pooling still looked quite similar at normal viewing size. Factor 4 was different. Since a larger area of pixels had to be represented by a single output pixel, the characteristics of each method became much more visible.

   With max pooling, brighter parts of the image stood out the most. This makes sense because the method always chooses the highest pixel value from each block. The effect was especially noticeable in the edges & text image, around the glowing traffic light, street signs, & other bright areas. Compared to other methods, some edges also appeared a bit stronger & harsher.

   Average pooling had a very different feel. Instead of keeping one particular pixel, it combines the values in a block by taking their average. The transition between areas therefore looked smoother, but that smoothness came with a trade off, details started to blur. The text on the street signs became softer & the effect was even easier to spot in the image full of leaves & branches, where many of the tiny details blended together.

   The median pooling result sat somewhere in between. Since it takes the median value rather than the brightest one/the average, unusually bright/dark pixels did not affect it as strongly. Visually, it gave a more balanced result than max pooling in the images I tested. Still, because the resolution itself had already been reduced, some fine information was naturally lost.

   So, while all 3 methods made the image smaller, they did not lose information in the exact same way. Max emphasized stronger values, average gave a softer look, & median was less sensitive to extreme values.

3. **Upsampling analysis**

For this part, all 3 methods started from the exact same low resolution image. This was important because I wanted the difference to come from the upsampling method, not from using different downsampled inputs.

The easiest method to recognize was nearest neighbor. When the image was enlarged, the pixels became much more obvious & some edges looked blocky. It was not always super noticeable when looking at the whole image, but once I zoomed into the Ludlow st sign, the stair like edges around the letters were much easier to see. Basically, the method makes the image larger by reusing the nearest available pixel rather than creating a gradual transition.

Bilinear interpolation removed a lot of that blocky appearance. Because the new pixel values are estimated from nearby pixels, transitions looked noticeably smoother. The downside is that this also softened the image, so some areas looked more blurry than with nearest neighbor.

Then there was bicubic interpolation. It also produced a smooth image, but some edges looked a little clearer compared to bilinear. The difference was not dramatic tho, when i looked at the full images, bilinear and bicubic were actually pretty hard to tell apart. Zooming into smaller details made the difference easier to notice.

The most interesting part for me was comparing all of them back to the original. Even after the images were enlarged to almost the same dimensions, none of them actually returned to the original quality. The size came back, but the missing detail did not.

4. **Overall observation**

What stood out the most from this experiment is that image size & image detail are not the same thing. Downsampling removes information & once that information is gone, simply enlarging the image again can't magically recreate it.

There also was not one method that looked "best" in every situation. Max pooling made strong & bright features stand out, average pooling created smoother results, while median pooling was less influenced by extreme values. On the upsampling side, nearest neighbor was clearly the most pixelated, bilinear traded some sharpness for smoothness, and bicubic gave a slightly clearer looking smooth result.

So in the end, choosing a method really depends on what we care about the image. Sometimes keeping strong features matters more, while in other cases a smoother result/cleaner edges might be more useful.